Optimal Control for Space Systems

EN.530.626 (Fall 2026)

Lecture 19

Instructor: Prof. Abhishek Cauligi
Course Assistant 1: Marvat Johri
Course Assistant 2: Patrick Schwartz

Class logistics

- Quiz 4 on Thursday 11/05 in-class
  - Updated guest lecture slides to Canvas

Agenda

- Last time: conic optimization
- Today: derivative-free optimization

Recall

In this class, we'll broadly study the following categories of optimization problems:

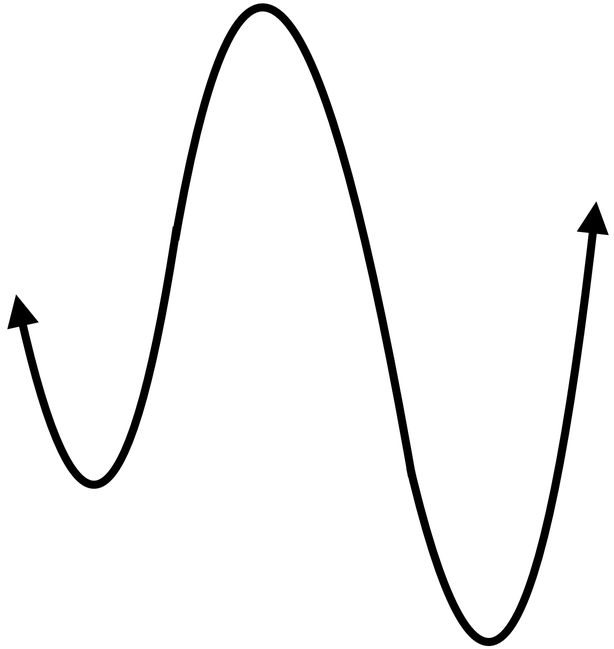
- Unconstrained vs. constrained
- Smooth vs. non-smooth
- Continuous vs. discrete
- Deterministic vs. stochastic
- Convex vs. non-convex

Recall

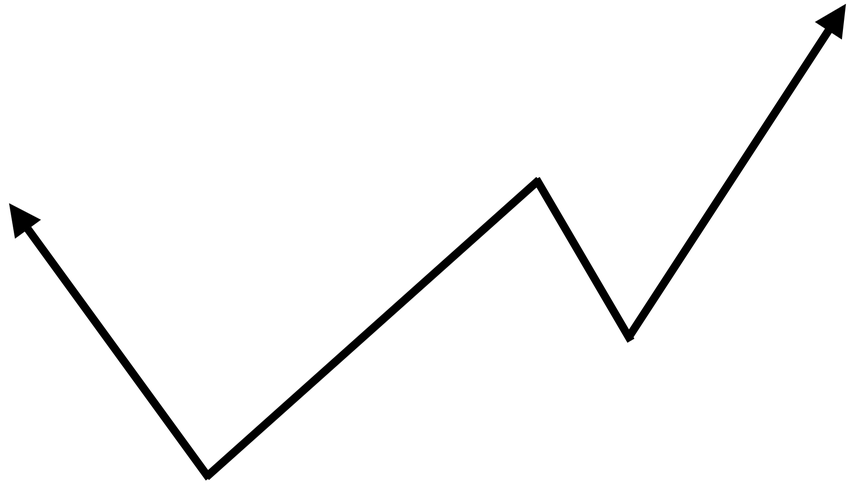
- Gradient-free optimization has gotten us far in class
- First/second-order methods can be applied to a broad class of problems
- Yet —a few shortcomings
  - What if the objective function or constraints are noisy?
  - What if the problem is *very* non-convex with multiple local minima?
  - What if computing the gradient or Hessian is intractable?

Stochastic methods

- Often, injecting noise into the optimization procedure yields performance



Smooth



Non-smooth

Cross-entropy method

- Iterative derivative-free method for optimization
- Maintains a *proposal distribution* to generate new samples for each iteration
- Tracks a number of *elite* samples to compute proposal distribution

Cross-entropy method

1. Generate an initial proposal distribution $\mathcal{N}(\mu^{(0)}, \Sigma^{(0)})$
2. Generate N_s samples $\{x^{(i)}\}_{i=1}^{N_s}$ and evaluate f for each of these
3. Using top N_{elite} samples re-compute

$$\mu^{(k+1)} = \frac{1}{N_{\text{elite}}} \sum_{i=1}^{N_{\text{elite}}} x^{(i)}$$

$$\Sigma^{(k+1)} = \frac{1}{N_{\text{elite}}} \sum_{i=1}^{N_{\text{elite}}} (x^{(i)} - \mu^{(i)})(x^{(i)} - \mu^{(i)})^T$$

Particle swarm optimization

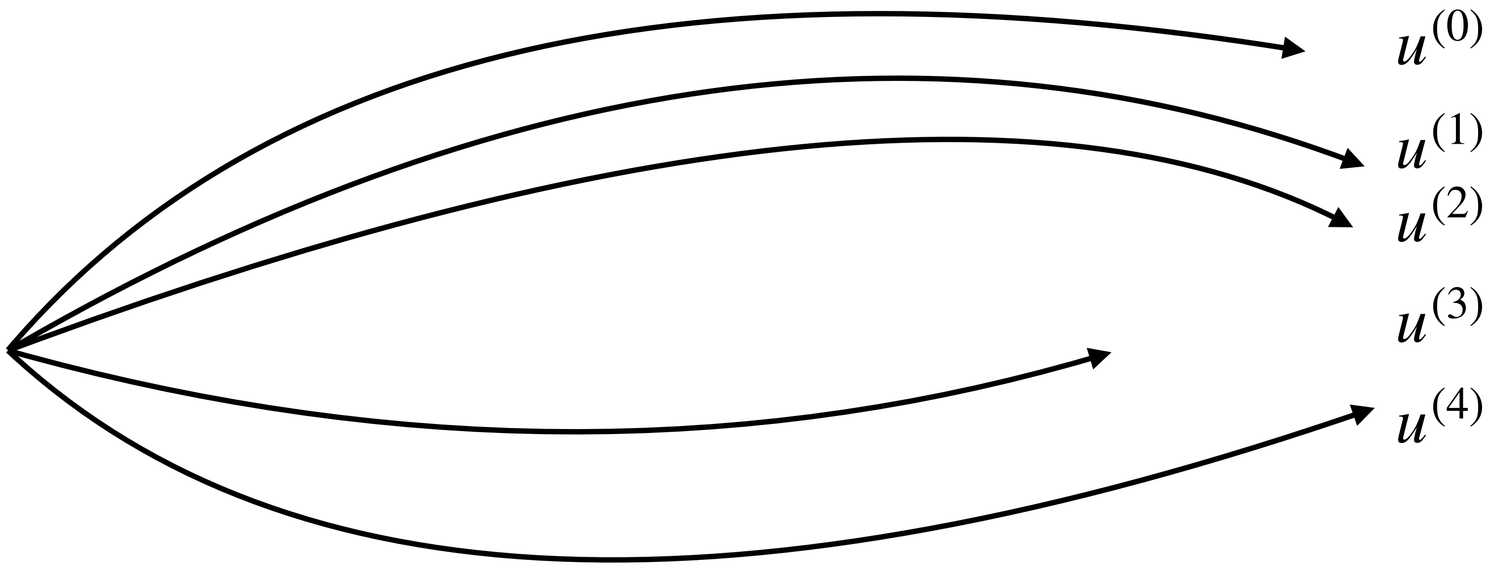
- Introduces notion of *momentum* to particles
- Track both the particles and the *best* version of that particle we have seen

Particle swarm optimization

1. Generate N_s samples $\{x^{(i)}\}_{i=1}^{N_s}$ and evaluate f for each of these
2. Compute velocity for each particle
   $v^{(i)} = wv^{(i)} + c_1 r_1 \cdot (x^{(i)}_{\text{best}} - x^{(i)}) + c_2 r_2 \cdot (x_{\text{best}} - x^{(i)})$
3. Update each particle $x^{(i)} = x^{(i)} + v^{(i)}$
4. Update $x^{(i)}_{\text{best}}$ if a particle i has improved its value

How to apply to control?

- Draw samples of controls and run derivative-free method of choice



MPPI

2016 IEEE International Conference on Robotics and Automation (ICRA)
Stockholm, Sweden, May 16-21, 2016

Aggressive Driving with Model Predictive Path Integral Control

Grady Williams, Paul Drews, Brian Goldfain, James M. Rehg, and Evangelos A. Theodorou

***Abstract*— In this paper we present a model predictive control algorithm designed for optimizing non-linear systems subject to complex cost criteria. The algorithm is based on a stochastic optimal control framework using a fundamental relationship between the information theoretic notions of free energy and relative entropy. The optimal controls in this setting take the form of a path integral, which we approximate using an efficient importance sampling scheme. We experimentally verify the algorithm by implementing it on a Graphics Processing Unit (GPU) and apply it to the problem of controlling a fifth-scale Auto-Rally vehicle in an aggressive driving task.**

and knowing the constraint requires precise knowledge of the system dynamics in the form of the coefficient of friction.

Another approach, which we consider here, is the optimal control approach which finds a sequence of controls or a control law that is optimal with respect to a given cost and system dynamics. This approach combines planning and execution into one phase by explicitly taking the dynamics into account during optimization. The problem with the optimal control approach is that, for systems with non-linear dynamics and costs, solving the optimal control problem is

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Check for updates

Model Predictive Path Integral Control: From Theory to Parallel Computation

Grady Williams,* Andrew Aldrich,† and Evangelos A. Theodorou‡
Georgia Institute of Technology, Atlanta, Georgia 30332

DOI: 10.2514/1.G001921

In this paper, a model predictive path integral control algorithm based on a generalized importance sampling scheme is developed and parallel optimization via sampling is performed using a graphics processing unit. The proposed generalized importance sampling scheme allows for changes in the drift and diffusion terms of stochastic diffusion processes and plays a significant role in the performance of the model predictive control algorithm. The proposed algorithm is compared in simulation with a model predictive control version of differential dynamic programming on nonlinear systems. Finally, the proposed algorithm is applied on multiple vehicles for the task of navigating through a cluttered environment. The current simulations illustrate the efficiency and robustness of the proposed approach and demonstrate the advantages of computational frameworks that incorporate concepts from statistical physics, control theory, and parallelization against more traditional approaches of optimal control theory.

MPPI

- Easy to implement and handles non-convex problems
- Easy to parallelize for fast implementation

Algorithm 1: Model predictive path integral control

Given: K: Number of samples
N: Number of time steps
$(\boldsymbol{u}_0, \boldsymbol{u}_1, \ldots \boldsymbol{u}_{N-1})$: Initial control sequence
Δt, $\boldsymbol{x}_{t_0}$, f, G, B, v: System/sampling dynamics
ϕ, q, R, λ: Cost parameters
$\boldsymbol{u}_{\text{init}}$: Value to initialize new controls to
while task not completed, **do**
  Generate random control variations $\delta \boldsymbol{u}$
  for $k \leftarrow 0$ **to** $K-1$, **do**
    $\boldsymbol{x} = \boldsymbol{x}_{t_0}$
    for $i \leftarrow 1$ **to** $N-1$, **do**
      $\boldsymbol{x}_{i+1} = \boldsymbol{x}_i + (f + G(\boldsymbol{u}_i + \delta \boldsymbol{u}_{i,k}))\Delta t$
      $\tilde{S}(\tau_{i+1,k}) = \tilde{S}(\tau_{i,k}) + \tilde{q}$
  for $i \leftarrow 0$ **to** $N-1$, **do**
    $\boldsymbol{u}_i \leftarrow \boldsymbol{u}_i + \left[\sum_{k=1}^{K} (\exp(-(1/\lambda)\tilde{S}_{(\tau_{i,k})}) \delta \boldsymbol{u}_{i,k} / \sum_{k=1}^{K} \exp(-(1/\lambda)\tilde{S}_{(\tau_{i,k})}))\right]$
  send to actuators $(\boldsymbol{u}_0)$
  for $i \leftarrow 0$ **to** $N-2$, **do**
    $\boldsymbol{u}_i = \boldsymbol{u}_{i+1}$
  $\boldsymbol{u}_{N-1} = \boldsymbol{u}_{\text{init}}$
  Update the current state after receiving feedback
  Check for task completion

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M. Gandhi, B. Vlahov, et al., “Robust model predictive path integral control: Analysis and performance guarantees", *IEEE Robotics and Automation Letters*.

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- C++ API
  - Class Hierarchy
  - File Hierarchy
  - Full C++ API
- Standard Documents
  - LICENSE
  - PACKAGE
  - README

README

Model Predictive Path Integral Controller

Overview

https://docs.ros.org/en/iron/p/nav2_mppi_controller/

Discussion

- What convergence guarantees exist?
- Can we guarantee constraint satisfaction?
- Can we combine ideas from derivative-free and gradient-based optimization?

Takeaways

- Derivative-free methods are often effective at:
  - Getting unstuck from local minima
  - Searching for global minima
  - Being easy to parallelize

References

- A. Jordana, J. Zhang, J. Amigo, and L. Righetti, “An Introduction to Zero-Order Optimization Techniques for Robotics”, 2026.